Checkpoint 6 · Testing & Fixing Matrix

Usability Testing Logs Across 6 Participants & 4 Core Fixes | INT335 Design Thinking

1. Usability Testing Protocol & Participant Sessions

Test #	Participant & Role	Task Assigned	Outcome	Observed Friction / Usability Hitch
T1	Student (Hosteller)	Report lost Casio calculator	Passed (48s)	Confused between 'Item Description' vs 'Secret Proof'.
T2	Day Scholar (Commuter)	Browse campus found feed	Passed (30s)	Filter dropdowns were too small on mobile screen.
T3	Security Guard (Block 34)	Verify claim & release item	Passed with aid	Needed Punjabi/Hindi labels; English was intimidating.
T4	Hosteller (Female student)	Check match notifications	Passed (22s)	Appreciated zero contact sharing; asked for safe pickup point.
T5	Library Staff (Desk Supervisor)	Log batch of 3 lost glasses	Passed (95s)	Requested bulk category logging for library closing hours.
T6	iPhone User (Mobile Chrome)	Change language in navbar	Failed initially	Language modal was cut off at top under Dynamic Island.

2. Feedback-to-Iteration Matrix (The 4 Critical Fixes)

Hitch Identified	Root Cause Analysis	Engineering Fix Implemented	Re-Test Validation
Fix 1: Language Modal Mobile Cutoff	Sticky navigation bar & iOS WebKit safe-area inset clipping modal container.	Implemented React `createPortal` rendering to body with `env(safe-area-inset-top)` padding.	100% visible on iPhone & Android with zero overlapping.
Fix 2: Language Accessibility for Staff	Security guards struggled with complex English technical terminology.	Added 7 Indian regional languages (Hindi, Punjabi, Telugu, Tamil, Bengali, Marathi, English).	Guards successfully verified claim in Punjabi in under 20s.
Fix 3: Secret Proof Confusion	Students accidentally typed public descriptions into the secret proof field.	Added explanatory tooltips & visual lock icon: 'Only campus security can see this'.	100% of participants entered private proofs (serial numbers, engraving).
Fix 4: NotificationBell Import Error	Vite import analysis failed due to missing `TranslatedText` component.	Created unified `TranslatedText.jsx` wrapper with fallback string support.	Production build compiled with 0 errors (`vite build` passed).